

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
I _{DDA(SCALER)}	Scaler static consumption from V _{DDA}	BRG_EN=0 (bridge disable)	-	0.2	0.3	μA
		BRG_EN=1 (bridge enable)	-	0.82	1	
t _{START_SCALER}	Scaler startup time	-	-	140	250	μs
t _{START}	Comparator startup time to reach propagation delay specification	High-speed mode	-	2	5	μs
		Medium mode	-	5	20	
t _D ⁽²⁾	Propagation delay for 200 mV step with 100 mV overdrive	High-speed mode	-	50	80	ns
		Medium mode	-	0.5	0.9	μs
	Propagation delay for step > 200 mV with 100 mV overdrive only on positive inputs	High-speed mode	-	50	120	ns
		Medium mode	-	0.5	1.2	μs
V _{offset}	Comparator offset error	Full common mode range	-	-	±20	mV
V _{hys}	Comparator hysteresis	No hysteresis	-	0	-	mV
		Low hysteresis	-	10	-	
		Medium hysteresis	-	20	-	
		High hysteresis	-	30	-	
I _{DDA(COMP)}	Comparator consumption from V _{DDA}	Medium mode	Static	-	5	μA
			With 50 kHz ±100 mV overdrive square signal	-	6	
		High-speed mode	Static	-	70	
			With 50 kHz ±100 mV overdrive square signal	-	75	

1. Refer to [Section 5.3.5: Embedded voltage reference](#).
2. Evaluated by characterization - Not tested in production.

5.3.21 Timer characteristics

The parameters given in [Section 5.3.21](#), [Table 61](#), and [Section 5.3.21](#) are specified by design, not tested in production.

Refer to [Table 47. I/O static characteristics](#) for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

Table 60. TIMx characteristics

Symbol	Parameter	Conditions	Min	Max	Unit ⁽¹⁾
t _{res(TIM)}	Timer resolution time	-	1	-	t _{TIMxCLK}
		f _{TIMxCLK} = 144 MHz	6.9	-	ns
f _{EXT}	Timer external clock frequency on CH1 to CH4	-	0	f _{TIMxCLK} /2	MHz
		f _{TIMxCLK} = 144 MHz	0	72	
Res _{TIM}	Timer resolution	TIMx (except TIM2/TIM5)	-	16	bit
		TIM2/TIM5	-	32	
t _{COUNTER}	16-bit counter clock period	-	1	65536	t _{TIMxCLK}
		f _{TIMxCLK} = 144 MHz	0.007	455.1	μs
t _{MAX_COUNT}	Maximum possible count with 32-bit counter	-	-	4, 294, 967, 296	t _{TIMxCLK}
		f _{TIMxCLK} = 144 MHz	-	29.826	s

1. TIMx, is used as a general term in which x stands for 1, 2, 5, 6, 7, 8, 12, 15, 16, 17.